

Machine learning researcher and engineer with 8+ years of experience across industry and academia, specializing in signal processing, machine learning, generative AI, and optimization. Experienced in developing and deploying machine learning solutions for large-scale ride-hailing operations, speech technology, and signal processing systems.

Work and Research Experience

7/2021– **Postdoctoral Fellow / PhD Student**, School of Electrical Engineering and Telecommunications, UNSW, Sydney.

Health-Related Speech Timing Analysis

- Investigating methods for extracting and analyzing speech timing markers to characterize cognitive, affective, mental, and physical changes for speech-based assessment and diagnosis.
- Developed a generalizable method that incorporates prosodic and timing-related acoustic cues (e.g., pitch) into the positional embeddings of transformer-based speech models, improving performance on tasks such as tone recognition and lexical stress detection.

Biologically Inspired Speech Signal Processing and Sequence Modeling

- Applied generative modeling techniques (e.g., VAEs, diffusion models) to model biologically inspired mechanisms for speech signal processing (e.g., speech recognition), bridging the gap between artificial technologies and human cognitive capabilities. Generalized these approaches to sequence modeling applications in other domains (e.g., household energy usage forecasting, time-series generation).
- Developed a variational CTC framework for order-preserving sequence modeling, leveraging both the strength of CTC and that of variational modeling, designed to mitigate certain types of errors.
- Proposed a relational-thinking-based acoustic modeling framework that captures relational information across domains (time and frequency) and scales (phone and word), thereby enhancing recognition performance.

Speech Data Annotation and Analysis Platform

- Developed a no-code speech data annotation and analysis platform that enables researchers to build machine learning-based pre-annotation pipelines on a visual canvas and efficiently produce high-quality annotated speech data through human-in-the-loop refinement.
- Integrated configurable modules (e.g., voice activity detection, speaker diarization, speech recognition) with multiple model options and an LLM-powered assistant for creating and refining speech pre-annotation pipelines.
- Contributed to the AusKidTalk project, which develops a large-scale annotated database of Australian children's speech to support child-focused speech recognition and speech development research.

6/2018–8/2021 **Algorithm Engineer**, Ride-Hailing Business Unit, Meituan, Beijing.

Machine Learning and Optimization for Ride-Hailing Operations

- Built and deployed end-to-end machine learning and optimization solutions for ride-hailing operations, covering driver order-volume forecasting, driver incentive allocation, cancellation responsibility detection, and driver/passenger feedback text classification. Managed the full lifecycle from feature engineering and model/solver development to A/B testing and production monitoring.
- Designed driver incentive strategies by combining driver response modeling, linear programming, and genetic algorithms, achieving up to 30% improvement in ROI over the baseline strategy.
- Developed and deployed an end-to-end cancellation responsibility detection pipeline, including ASR transcription of driver-passenger calls, integration of contextual features (e.g., pickup distance, POI category), conversational text classification, and downstream enforcement; achieved at least 95% accuracy in detecting driver-induced passenger cancellations.
- Built and deployed a multi-label classifier for driver and passenger feedback texts, using syntactic-structure-based data mining for sample collection and achieving at least 90% precision and 80% recall.

7/2015–6/2018 **Research & Development Engineer**, Radar and Electronic Equipment Research Institute, Aviation Industry Corporation of China (AVIC), Beijing.

Emitter Signal Recognition

- Developed statistical and machine learning/deep learning algorithms (e.g., SVMs, CNNs) for radar waveform and intra-pulse modulation recognition using spectro-temporal and pulse features.
- Applied multivariate hypothesis testing to carrier frequency, pulse width, pulse repetition interval, and direction-of-arrival features to identify radar emitters and distinguish between similar emitter models.
- Developed clustering methods to infer radar operating modes (search and track) from amplitude sequences, and delivered supporting algorithm libraries for airborne electronic warfare systems.

9/2012–6/2015 **Postgraduate Student**, School of Information Science and Engineering, Fudan University, Shanghai.

Energy-Efficient Transmission for Energy-Harvesting Wireless Communications

- Designed optimization frameworks for energy-harvesting-powered point-to-point, MIMO uplink, and MIMO downlink channels, targeting throughput maximization and energy minimization under practical constraints such as battery capacity, transmission deadlines, and non-ideal circuit power consumption.
- Applied convex optimization theory to develop low-complexity algorithms for computing optimal transmission strategies under energy and latency constraints.
- Extended offline optimization solutions to online algorithms adaptable to practical wireless communication systems.

Education

7/2021–6/2025 **University of New South Wales**, Sydney, Australia, PhD, Signal Processing.

9/2012–6/2015 **Fudan University**, Shanghai, China, M.Eng.Sc., Communication and Information Systems.

9/2008–6/2012 **Beijing Institute of Technology**, Beijing, China, B.Eng., Communication Engineering.

Technical Skills

Signal Processing: speech recognition, speech annotation, probabilistic modeling, sequence modeling, time series analysis.

Machine Learning: machine learning, deep learning, transformer models, generative modeling, variational autoencoders, diffusion models.

Optimization: convex optimization, linear programming, PuLP, OR-Tools, CVX.

Programming: Python, PyTorch, TensorFlow, pandas, scikit-learn, Java, C++.

Publications

Google Scholar

Author or co-author of 20 papers in leading signal processing journals and conferences, including IEEE Transactions, ICASSP, INTERSPEECH, and ICC.

- [1] **Z. Nan**, T. Sirojan, V. Sethu, and B. Ahmed, “Interpolation-augmented diffusion for accurate high-frequency spectral modeling,” submitted to *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- [2] **Z. Nan**, T. Sirojan, V. Sethu, and B. Ahmed, “Phase distortion analysis of diffusion models,” submitted to *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- [3] **Z. Nan**, T. Sirojan, M. Shahin, T. Szalay, V. Sethu, and B. Ahmed, “SpeechBench: a unified speech annotation and analysis tool,” in *Proc. INTERSPEECH*, 2026.
- [4] T. Szalay, M. Shahin, T. Sirojan, **Z. Nan**, R. Huang, J. Arciuli, E. Baker, F. Cox, K. Ballard, and B. Ahmed, “AusKidTalk: developing an orthographic annotation workflow for a speech corpus of Australian English-speaking children,” *Language Resources and Evaluation*, vol. 60, no. 53, pp. 1–35, July 2026.
- [5] T. Szalay, **Z. Nan**, R. Huang, M. Shahin, T. Sirojan, K. Ballard, and B. Ahmed, “AusKidTalk: developing transcription guidelines for continuous Australian English child speech,” in *Proc. Language Resources and Evaluation Conference (LREC)*, 2026.
- [6] Y. Rong, P. Chen, and **Z. Nan**, “Underwater diver communication system based on speech/text conversion,” in *Proc. IEEE OCEANS*, 2025.
- [7] T. Szalay, M. Shahin, T. Sirojan, **Z. Nan**, R. Huang, K. Ballard, and B. Ahmed, “AusKidTalk: using strategic data collection and out-of-domain tools to semi-automate novel corpora annotation,” in *Proc. INTERSPEECH*, 2025.
- [8] **Z. Nan**, T. Dang, V. Sethu, and B. Ahmed, “A joint spectro-temporal relational thinking based acoustic modeling framework,” submitted to *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, arXiv:2409.15357.
- [9] **Z. Nan**, T. Dang, V. Sethu, and B. Ahmed, “Variational connectionist temporal classification for order-preserving sequence modeling,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
- [10] M. Shahin, **Z. Nan**, V. Sethu, and B. Ahmed, “Improving wav2vec2-based spoken language identification by learning phonological features,” in *Proc. INTERSPEECH*, 2023.
- [11] **Z. Nan**, T. Zhao, and T. Huang, “Cramer-Rao lower bounds of joint time delay and Doppler-stretch estimation with random stepped-frequency signals,” in *Proc. IEEE International Conference on Digital Signal Processing (DSP)*, Beijing, China, Oct. 2016.
- [12] T. Zhao, **Z. Nan**, and T. Huang, “Performance analysis of joint time delay and Doppler-stretch estimation with random stepped-frequency signals,” in *Proc. IEEE International Conference on Digital Signal Processing (DSP)*, Beijing, China, Oct. 2016.
- [13] **Z. Nan**, T. Chen, X. Wang, and W. Ni, “Energy-efficient transmission schedule for delay-limited bursty data arrivals under non-ideal circuit power consumption,” *IEEE Transactions on Vehicular Technology*, vol. 65, no. 8, pp. 6588–6600, Aug. 2016.

- [14] X. Wang, **Z. Nan**, and T. Chen, “Optimal MIMO broadcasting for energy harvesting transmitter with non-ideal circuit power consumption,” *IEEE Transactions on Wireless Communications*, vol. 14, no. 5, pp. 2500–2512, May 2015.
- [15] W. Li, **Z. Nan**, and X. Wang, “Optimal transmission policy for energy-harvesting powered MIMO multi-access channels,” in *Proc. IEEE/CIC International Conference on Communications in China (ICCC)*, Shanghai, China, Oct. 2014.
- [16] **Z. Nan**, T. Chen, and X. Wang, “Energy-efficient transmission of delay-limited bursty data packets over time-varying channels under non-ideal circuit power,” in *Proc. IEEE China Summit and International Conference on Signal and Information Processing (ChinaSIP)*, Xi’ an, China, Jul. 2014.
- [17] **Z. Nan** and X. Wang, “Energy-efficient transmission of delay-limited bursty data packets under non-ideal circuit power consumption,” in *Proc. IEEE International Conference on Communications (ICC)*, Sydney, Australia, Jun. 2014.
- [18] X. Wang and **Z. Nan**, “Optimal MIMO broadcasting for energy harvesting transmitter with non-ideal circuit power,” in *Proc. IEEE International Conference on Communications (ICC)*, Sydney, Australia, Jun. 2014.
- [19] **Z. Nan**, T. Chen, and X. Wang, “Optimal MIMO broadcasting over time-varying wireless channels for energy harvesting transmitter with non-ideal circuit power,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Florence, Italy, May 2014.
- [20] X. Wang and **Z. Nan**, “An optimization approach to energy-efficient transmissions with strict deadlines,” in *Proc. IEEE Wireless Communications and Networking Conference (WCNC)*, Shanghai, China, Apr. 2013.

Awards and Honors

- Australian Government Research Training Program (RTP) Scholarship.
- Second Prize, INTERSPEECH 2023 MERLion CCS Challenge.
- Project Star Award, Ride-Hailing Business Unit, Meituan.
- Second Prize, 5th Fudan University Doctoral Forum.